

Beyond Single-Target Grounding: Few-Shot Generalized Referring Expression Comprehension with Frozen Region-Aware Representations

Background

When a vision system is asked to locate *the child in a yellow raincoat*, the correct answer may be no object at all. Likewise, for an expression such as *the two cyclists on the right*, returning only one bounding box is incomplete. Most conventional referring expression comprehension (REC) settings avoid these cases by assuming that every expression refers to exactly one object. Generalized Referring Expression Comprehension (GREC) removes this assumption: an expression may refer to **zero, one, or multiple objects**, and the model must output an appropriately sized set of boxes (He et al., 2023). This project investigates whether such reliable behavior can be learned under limited supervision without training a large end-to-end grounding model.

Problem

The central question is: **under few-shot supervision, which frozen visual-language representations are most suitable for generalized grounding over no-target, single-target, and multi-target expressions?** This question is interesting for both practical and methodological reasons. A single-target model is forced to produce a box even when the queried object is absent, which is a serious failure mode in interactive vision systems. It also cannot represent queries that legitimately describe several objects. Existing GREC methods, including RECENTFormer and HieA2G, address variable numbers of targets through specialized architectures and explicit target-count reasoning (Hemanthage et al., 2024; Wang et al., 2025). However, these systems are relatively expensive to train and adapt because they rely on larger task-specific architectures and substantial supervision. In contrast, this project investigates whether generalized grounding can be achieved with a lightweight solution based on frozen pretrained representations and a small trainable prediction head. Such a design is attractive in few-shot settings because it reduces computational cost, lowers the amount of trainable parameters, and makes it easier to compare the intrinsic quality of different pretrained representations without the results being dominated by large-scale end-to-end optimization.

Solution

Rather than competing with these models through another large architecture, I propose a controlled study of a lower-cost alternative: **freezing pretrained representations and adapting only a compact, cardinality-aware prediction head**. This makes it possible to ask a focused question about representation quality rather than confounding it with extensive model training.

Related Work

The background reading will be organized around three parts. First, GREC and [gRefCOCO](#) provide the task definition, dataset, and evaluation setting for expressions with zero, one, or multiple referents (He et al., 2023). RECENTFormer and HieA2G will provide context on how recent supervised methods model variable target counts and complex cross-modal alignment (Hemanthage et al., 2024; Wang et al., 2025).

Second, I will examine frozen representation models that could support a lightweight grounding system. [CLIP](#) learns transferable image-text representations from large-scale natural language supervision (Radford et al., 2021), while ReCLIP shows that CLIP can be repurposed for region scoring in REC, but also reports that

CLIP alone is weak at spatial reasoning (Subramanian et al., 2022). [DINOv2](#) provides strong self-supervised visual features that may help distinguish visually similar candidate regions, although it does not itself supply text alignment (Oquab et al., 2024). [RegionCLIP](#) is relevant because it explicitly targets region-text alignment, addressing the mismatch between image-level pretraining and region-level recognition (Zhong et al., 2022). Most importantly, [SigLIP 2](#) reports improved localization and dense prediction capabilities through a training recipe that supplements image-text alignment with captioning-based and self-supervised objectives (Tschannen et al., 2025). This makes it a natural main representation candidate for generalized grounding.

Finally, C-REC and FineCops-Ref will provide context for evaluating whether a model can reject fine-grained negative or counterfactual expressions instead of confidently grounding nonexistent targets (Yu and Li, 2024; Liu et al., 2024).

Dataset

The main dataset will be [gRefCOCO](#), which contains referring expressions associated with no target, a single target, or multiple targets over [MS COCO](#) images (He et al., 2023). I will use its existing train and evaluation splits rather than collecting new annotations. To create a few-shot setting, I will train on stratified subsets containing **1%, 5%, and 10%** of the training expressions, preserving the proportions of no-target, single-target, and multi-target cases as closely as possible. This is important because a randomly selected small subset could contain too few no-target or multi-target examples for the model to learn meaningful cardinality behavior. For an additional reliability test, I will evaluate the trained models on the counterfactual datasets C-RefCOCO, C-RefCOCO+, and C-RefCOCOg introduced by Yu and Li (2024), provided that their preprocessing can be made consistent with the main pipeline. FineCops-Ref will be treated as an optional extension for testing more difficult compositional negative cases rather than as a required part of the project.

Proposed Method

I will formulate the task as **candidate-set selection**. Proposal-based reasoning has been widely used in referring expression comprehension: earlier models score candidate regions against language descriptions, while ReCLIP specifically repurposes CLIP to evaluate object proposals through region-level image-text matching (Yu et al., 2018; Subramanian et al., 2022). Following this general formulation, for each image and expression, a fixed pretrained region proposal model will generate a shared pool of candidate boxes. The proposal model will remain frozen and identical for every representation being compared. Each candidate will be associated with the ground-truth target set according to its intersection-over-union with annotated boxes during training, and the grounding model will predict a subset of the proposals at test time.

Unlike standard REC settings that assume exactly one valid output region, this project follows the generalized task definition introduced by GREC, in which an expression may correspond to zero, one, or multiple targets (He et al., 2023). The use of a fixed proposal pool is a deliberate experimental choice: it keeps the comparison focused on **visual-language representation and cardinality-aware selection rather than on detector training**. Because a proposal-based system cannot recover objects missing from its candidate pool, I will additionally report proposal recall at an IoU threshold of 0.5 as an upper-bound diagnostic for the grounding stage.

The proposed model consists of a frozen representation encoder and a lightweight trainable prediction head. I will compare four representation settings:

1. **CLIP region-text matching**, serving as the basic frozen visual-language baseline. This setting is directly motivated by ReCLIP, which demonstrates that CLIP can score candidate regions for referring expression comprehension without task-specific training (Subramanian et al., 2022).
2. **CLIP with DINOv2 visual features**, testing whether self-supervised local visual features provide

complementary information for distinguishing visually similar candidates. The motivation for this comparison comes from evidence that CLIP-like representations can miss visually salient distinctions and may benefit from additional vision-only features, although this combination has not been established as a GREC method itself (Tong et al., 2024; Oquab et al., 2024).

3. **SigLIP 2**, serving as the main proposed representation because it reports stronger localization and dense prediction capabilities while retaining aligned image-text representations (Tschannen et al., 2025).
4. **RegionCLIP**, included if implementation time permits, as a stronger region-aligned comparison because it explicitly learns region-text representations rather than relying only on image-level language supervision (Zhong et al., 2022).

For every candidate box, the model will extract a region representation and combine it with the expression representation, a region-text similarity score, and normalized box coordinates. Incorporating location information is consistent with prior REC models such as MAttNet, which explicitly models location cues when resolving referring expressions (Yu et al., 2018). This design is also motivated by ReCLIP’s finding that CLIP-based matching alone is weak at spatial reasoning (Subramanian et al., 2022). A small multilayer perceptron will output a membership score for each candidate box:

$$p_i = P(b_i \in Y \mid I, T, B),$$

where Y denotes the ground-truth set of referred boxes.

A second pooled head will predict the number of referred objects, using count classes such as zero, one, two, and three-or-more, adjusted according to the training-set distribution. Explicit target-count modeling has already been shown to be relevant for generalized referring expression comprehension: HieA2G introduces an Adaptive Grounding Counter to determine the number of target objects dynamically (Wang et al., 2025). In this project, the purpose of the cardinality head is not to claim a new counting mechanism, but to provide **a compact and uniform output module for comparing frozen representations under few-shot supervision**.

At inference time, a predicted count of zero produces an empty set; otherwise, the system selects the highest-scoring boxes according to the predicted count. The training objective will combine binary box-membership loss with a cardinality classification loss:

$$\mathcal{L} = \mathcal{L}_{\text{membership}} + \lambda \mathcal{L}_{\text{cardinality}}.$$

All encoder weights will remain frozen, and the same proposal pool, head structure, optimization schedule, and few-shot samples will be used across representation variants. This controlled design distinguishes the project from prior task-specific GREC architectures: rather than proposing another large end-to-end grounding model, it studies whether **modern frozen region-aware representations can support reliable variable-cardinality grounding through a small shared adapter**.

This framework enables several controlled comparisons. Comparing CLIP region matching with CLIP plus the learned cardinality-aware head will measure the importance of explicitly modeling variable-size outputs. Comparing CLIP with CLIP+DINOv2 will test whether additional self-supervised visual information improves generalized grounding under limited supervision. Comparing CLIP+DINOv2 with SigLIP 2 will test whether a single, more localization-oriented visual-language encoder is more effective than manually combining separate semantic and visual features. If RegionCLIP is included, it will further indicate whether explicit region-text pretraining remains advantageous over a newer general-purpose visual-language representation.

Evaluation

The primary evaluation will follow the released [GREC evaluation protocol on gRefCOCO](#). In addition to the overall generalized grounding score, I will report results separately for no-target, single-target, and multi-target expressions. The most informative supplementary measures will be no-target accuracy, false grounding rate on no-target expressions, single-target localization accuracy, multi-target set F1 or exact-set accuracy, and cardinality accuracy. I will also compare performance across the 1%, 5%, and 10% supervision settings and report results over multiple random few-shot samples where feasible. For the counterfactual evaluation, the main question will be whether models trained for generalized grounding can **reject expressions in which a crucial attribute has been changed so that no valid referent remains**.

Expected Results

I expect a simple CLIP-based matcher to remain a reasonable baseline on ordinary single-target cases, but to **fail more often on no-target and multi-target expressions** because it lacks an explicit mechanism for returning an empty or variable-sized set. Adding a cardinality-aware head should substantially reduce forced grounding errors. DINOv2 features may help when targets differ through local appearance cues, while SigLIP 2 may offer a stronger balance of text alignment and regional localization. However, the purpose of the project is not to assume that SigLIP 2 will dominate every setting. A negative result would also be informative: if a region-specialized representation such as RegionCLIP remains stronger, or if representation improvements do not transfer to counterfactual rejection, this would show that dense localization quality alone is not sufficient for reliable generalized grounding.

The contribution of this project is therefore not a new GREC benchmark or a new large grounding architecture. It is **a controlled few-shot study of how much reliable variable-target grounding can be obtained from frozen representations and a small cardinality-aware adapter**. This is a manageable experimental scope, while still addressing a failure mode that standard single-target grounding formulations leave unresolved.

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